AYUSH SINGH

SUMMARY

Software Engineer III (L4) at Google (Munich) with over **4 years** of experience architecting planet-scale distributed systems, low-latency **C++, Java, and Go** microservices, fault-tolerant storage systems, and real-time streaming pipelines. Standardized production architectures across **5,200+ server nodes**, scaled internal storage backing **100,000+ services**, and delivered **$2.5M+ in quarterly savings**. Earned a fast-track promotion to L4 in 15 months, a **VP of Engineering Spot Bonus**, and **3 Google Readability certifications** across **1,040+ code reviews**.

SKILLS

Languages

C++ · Java · Go · TypeScript · Python · Scala · Kotlin · SQL

Distributed Systems & RPC

gRPC · Protobuf · HDFS / FUSE · Paxos / Raft · Apache Spark · Kafka · ActiveMQ

Storage, Cloud & Observability

Spanner · Snowflake · Redis · PostgreSQL · Kubernetes · Docker · OpenTelemetry · GCP / AWS · Terraform

AWARDS

- **VP of Engineering Spot Bonus** & **7 Peer Bonuses** at Google for global launches and storage reliability.
- **3 Google Readability Certifications** in C++, TypeScript, and API Design (**1,040+ code reviews**).
- **Rank #1 in Institute & #72 Worldwide** on GeeksforGeeks (top **0.07%** of 100,000+ developers).
- **Indian Academy of Sciences Fellow** (**1 of 11** nationwide) & **Google ML Facilitator** (**1 of 50**).

FIND ME ONLINE

GitHub
https://github.com/iu-sh

Portfolio & Interactive Resume
https://ayushsingh.is-a.dev

LinkedIn
https://linkedin.com/in/iu-sh

Software Engineer III (L4) · Distributed Systems & Core Infrastructure

+49 1735 34 8915 apparently.ayush@gmail.com
https://linkedin.com/in/iu-sh Munich, Germany

EXPERIENCE

Software Engineer III
11/2024 - Present
Google — Munich, Germany

- Architected high-throughput, low-latency request processors within Google's core Java backend framework and C++ Search serving engine, powering over **2 billion daily requests** at **99.99% availability**.
- Standardized the organization-wide microservice architecture defaulted across **5,200+ server nodes** and **210+ production modules** (Search, YouTube, Maps), reducing signal-processing complexity by **40%** and eliminating **1,300+ lines of code**.
- Implemented a parallel stream validation pipeline across YouTube, Play, Search, and Payments, verifying **100% live migration parity** across **10 billion+ daily RPCs** while migrating **7 identity pipelines** to batch compute.
- Spearheaded end-to-end backend launches for European Online Safety and US App Store regulatory compliance across Web, Android, and iOS within **4 weeks** across **5 engineering teams** (**VP of Engineering Spot Bonus**).

Software Engineer II
07/2022 - 11/2024
Google — Hyderabad, India

- Engineered fault-tolerant storage and replication components for Google's internal distributed file system, accelerating binary delivery across **100,000+ production services** and reducing tail latency by **28%**.
- Deployed server-side admission control, bounded semaphore RPC concurrency limits in the **FUSE client daemon**, and decentralized **Paxos snapshot locks**, cutting lock contention by **65%** during peak cluster load.
- Designed an automated service-level compliance and observability platform correlating traces and time-series metrics across **40+ product teams**, generating **$2.5M+ in quarterly infrastructure savings**.
- Co-developed a real-time **OpenTelemetry** trace collector over gRPC and an encrypted **Go asset gateway** handling **50,000+ requests per second**, achieving a **fast-track promotion to L4 in 15 months**.

Software Engineer Intern
01/2022 - 06/2022
Komprise — Bengaluru, India

- Automated multi-threaded storage verification across parallel cluster nodes, cutting build verification runtime by **87.5%** and increasing critical bug detection coverage by **60%**.

Data Engineer Intern
05/2021 - 07/2021
Accenture — Bengaluru, India

- Constructed high-volume cloud data pipelines transferring **10TB+ datasets** from **AWS S3 to Snowflake Data Warehouse**, reducing manual analytics reporting time by **70%**.

Software Engineer Intern
10/2020 - 01/2021
Tookitaki — Singapore

- Optimized a real-time **Apache Spark Structured Streaming and ActiveMQ** pipeline with connection pooling and fault-tolerant recovery, reducing end-to-end streaming latency by **94%** from **850ms to 45ms**.

PROJECTS

YTM AI Host (Real-Time AI Radio Jockey)
github.com/iu-sh/ytm_ai_host
TypeScript, Python, Gemini API, WebGPU

- Developed a local audio streaming server and browser extension for YouTube Music and Spotify combining hardware-accelerated GPU inference and neural TTS for **sub-100ms audio transitions** across **1,000+ tracks**.

LinkinJMS & Incident Triage Agents
github.com/iu-sh/linkinJMS
Scala, Apache Spark, ActiveMQ, Python

- Created an open-source bidirectional ActiveMQ streaming connector for Apache Spark sustaining **sub-50ms ingestion latency** across **100,000+ messages/sec**, plus a production multi-agent root-cause triage system.

EDUCATION

Bachelor of Technology in Electrical Engineering
08/2018 - 05/2022
National Institute of Technology, Agartala — CGPA: 9.32 / 10.0